Graduation Project – Supply Chain

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Source

My Superstore.xlsx

Superstore Data Analysis Summary

- **The Data Structure:** The file connects **Transactional Data** (Orders, Shipping, and Profits) with **Strategic KPIs** (Service Level Agreements).
- **Operational Core:** It tracks the efficiency of the supply chain through metrics like **"Days to Ship"** and **"Shipping Modes"** across global regions.
- **Performance Benchmarking:** The second sheet defines the **SLA targets** (e.g., 92% for on-time delivery). My analysis focuses on measuring the **Gap** between actual fulfillment and these targets.
- **Key Objective:** To identify logistical bottlenecks and regional performance variances to improve overall **Order Fulfillment Rate**.

Cleaning Data

Cleaning Superstore Data

clean super store.ipynb

1. **Missing Values Imputation:**
   - Identified null values in the **Subregion** column.
   - Mapped and filled these missing values using a logical link between **State**, **Country**, and their corresponding **Subregions** to ensure geographical data integrity.
2. **SLA Logic Engineering:**
   - Created a status column named **"Achieved"** by comparing the **Ship Date** against the **Due Date**.
   - Categorized each order as **"achieved"** (if shipped on or before the deadline) or **"failed"** (if shipped after the deadline).
3. **Key Performance Indicator (KPI) Extraction:**
   - Calculated the **Service Level %** by measuring the ratio of on-time deliveries ("achieved") against the total number of orders.
4. **Temporal Consistency:**
   - Ensured all date-related fields were correctly formatted to perform accurate time-based calculations and logical comparisons.

clean
superstore.xlsx

Tools

Tool	Task
Cleaning	Python
Visualization	Power BI

Data Key Performance and Calculating fields

Project Objective

This project evaluates supply chain efficiency by analyzing order fulfillment performance, focusing on delivery timeliness and service level using OTIF (On-Time In-Full).

Business Problem

Organizations face challenges in delivering orders on time, leading to reduced customer satisfaction and operational inefficiencies.

Dataset Metadata – Superstore

Order Information

- Order ID: Unique identifier for each order
- Order Date: Date when order was placed
- Ship Date: Date when order was shipped
- Due Date: Expected delivery date

Customer Information

- Customer Name: Name of the customer
- Segment: Customer category

Location Information

- Region: Geographic region
- State / City: Location details

Shipping Information

- Ship Mode: Shipping method

Product Information

- Quantity: Number of items ordered

Financial Data

- Sales: Revenue
- Profit: Profit value

KPIs

- Service Level (%) = OTIF / Total Orders

- On-Time %
- Average Delivery Time
- Average Delay
- Total Orders

Analysis Questions

- What is the service level?
- Which customers have the lowest service level?
- Which regions show delays?
- How does shipping mode affect performance?
- Is performance improving over time?

Dashboard Overview

- Page 1: KPIs and overall performance
- Page 2: Customer and region analysis
- Page 3: Logistics and delivery performance

Key Insights

- Service level varies across regions
- Delivery delays impact performance
- Some customers are more affected

project.pbix

Data Visualization

1. Key Performance Indicators (KPIs)

- **Total Orders:** Count of all customer orders processed.
- **On-Time %:** Percentage of orders delivered within the promised time frame.
- **OTIF %:** Percentage of orders delivered both on time and in full quantity.
- **Fill Rate %:** Percentage of requested quantities that were successfully fulfilled from available stock.
- **Service Level:** Overall measure of how well customer demand was met, reflecting reliability of supply and delivery.

2.Charts Calculated

- **Regional Comparison:** A bar chart compares Fill Rate, OTIF, and Service Level across regions (EMEA, Latam, AsiaPac, North America).
- **Performance Insight:** Highlights which region achieved the strongest service levels and which region lagged, mainly due to lower fulfillment performance.

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